Chiyoda: Entropy-Guided Information Control for Hazard-Coupled Pedestrian Evacuation

A Simulation Framework for Measuring When Emergency Communication Helps or Harms

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Abstract

Emergency evacuation is usually modeled as a physical routing problem, with information appearing as a fixed assumption. We present *Chiyoda*, an Information-Theoretic Evacuation Dynamics framework that treats emergency communication as a controllable intervention in a coupled physical-information system. Agents carry probabilistic beliefs, exchange distorted information through local gossip, receive beacon and responder messages, suffer physiological impairment under hazards, and choose routes through belief-weighted path planning. The framework compares static, global, responder-relay, entropy-targeted, density-aware, exposure-aware, and bottleneck-avoidance policies. The central claim is not that more information is always better, but that emergency communication has measurable safety trade-offs.

1 Introduction

Emergency communication is usually treated as a benign input to evacuation: give people better information and their decisions should improve. That assumption is plausible at the individual level, but it is not obviously true at the crowd level. A warning that reduces uncertainty for everyone at once can also synchronize route choice, push agents toward the same exit, amplify a bottleneck, or move people through a contaminated region before the physical hazard has dissipated. The question is therefore not simply whether information helps, but when, where, and for whom it helps.

This paper studies evacuation as a coupled physical-information system. The physical side includes layout geometry, social-force movement, bottleneck dynamics, hazard spread, and physiological impairment. The information side includes partial exit knowledge, hazard beliefs, signage and PA broadcasts, responder credibility, local gossip, message decay, and distortion. Chiyoda connects these layers by making agents plan through their believed world rather than the ground truth world. They may avoid a hazard they have heard about, miss an exit they do not know, herd when uncertainty is high, or follow a stale route because the belief has not yet decayed.

Research question. We ask whether entropy-guided communication can improve evacuation outcomes without creating harmful convergence. More concretely: which intervention policies reduce belief entropy and improve belief accuracy per unit of induced exposure, queueing, and exit imbalance? The current study scaffold compares static signage, global broadcasts, responder relay, entropy-targeted broadcasts, density-aware targeting, exposure-aware targeting, and bottleneck-avoidance targeting.

Contributions. This paper makes four contributions.

1. **A controllable information-intervention layer.** Chiyoda turns emergency communication into a first-class policy object rather than a fixed scenario assumption. Policies can be static, authority-driven, or adaptive to entropy, density, exposure, and bottleneck pressure.

2. **A coupled ITED simulation loop.** Agents carry probabilistic beliefs over exits and hazards, update them through observation, beacon broadcast, responder relay, and gossip, then route through belief-weighted costs under social-force dynamics and hazard impairment.

3. **Information-safety metrics.** We introduce study outputs that jointly measure entropy reduction, belief accuracy gain, intervention reach, hazard exposure pressure, queue pressure, exit imbalance, and a harmful convergence index.

4. **A reproducible paper pipeline.** YAML study definitions generate multi-seed bundles with Parquet/CSV tables and paper figures. The LaTeX paper ingests generated study statistics through `stats.tex`.

Roadmap. §2 positions the work against pedestrian dynamics, information-aware evacuation, and ABM validation work. §3 describes Chiyoda's model and intervention policies. §4 explains the software pipeline. §5 lays out the planned empirical study. §6 sharpens the distinction from prior work. §7 states the current threats to validity, and §8 concludes.

2 Background

Pedestrian evacuation models have a long physical tradition. The social-force model represents pedestrians as self-driven particles subject to destination, repulsion, and contact forces [2]. Cellular automata models discretize space and update local movement decisions through transition rules [1]. Fundamental-diagram work relates crowd density to walking speed and flow capacity [9]. These tools are useful because evacuation failure is often physical: bottlenecks, counter-flow, density waves, and slower-is-faster effects can dominate individual intent.

Hazard-coupled evacuation adds another layer. In CBRN or smoke scenarios, the environment changes while people move through it; visibility, speed, and physiology can degrade as a function of exposure. Prior toxic-gas evacuation models already show that information perception and transmission matter for route choice under a spreading contaminant [4, 3]. Chiyoda builds on this insight but makes the communication policy itself the experimental object.

Information-aware evacuation models also intersect with game theory and bounded rationality. Bayesian formulations can model incomplete information and route-choice equilibria under congestion [8]. Social group ABMs review the importance of communication, affiliation, and collective behavior, while also noting persistent gaps in behavioral realism and validation [7, 5]. Chiyoda takes a simulation-first position: it does not claim to solve empirical validation in one paper, but it makes belief uncertainty measurable and interventions reproducible.

The information-theoretic lens comes from Shannon entropy [6]. For each agent, Chiyoda computes entropy over exit existence, hazard knowledge, and general danger. Global entropy is the population mean. Entropy is not a normative objective by itself: the point of this paper is to test when reducing entropy improves safety and when it merely concentrates the crowd's behavior.

3 Design

Chiyoda models evacuation as a feedback loop between physical state, information state, and action. At each step, hazards evolve; agents observe nearby exits and hazards; beacons, responders, gossip, and intervention policies update beliefs; agents revise intentions; belief-weighted navigation selects paths; social-force dynamics move agents; and telemetry records the macroscopic outcome.

3.1 Belief State

Each agent carries a belief vector over exits, hazards, and general danger. Exit beliefs store existence probability, congestion estimate, freshness, source credibility, and gossip hop count. Hazard beliefs store position, severity estimate, radius estimate, freshness, credibility, and hop count. Agents do not route through the ground truth environment directly; they route through the subset of exits and hazards represented in their beliefs.

3.2 Communication Channels

The baseline ITED channels are direct observation, beacon broadcast, and agent-to-agent gossip. Observation is range-limited by visibility and physiological impairment. Beacons provide high-credibility exit knowledge within a fixed radius. Gossip is local, probabilistic, and distorted by hop count and panic state.

3.3 Intervention Policies

The new contribution is an intervention policy interface. Every policy decides whether to fire, selects one or more spatial targets, builds a message, applies it to recipients inside a radius, and emits telemetry. Chiyoda currently implements seven policies:

- **Static beacon**: periodic local messages from existing signage or PA beacon locations.
- **Global broadcast**: station-wide high-reach communication.
- **Responder relay**: mobile high-credibility messages emitted by released responders.
- **Entropy-targeted**: targets agents with highest belief entropy.
- **Density-aware**: targets dense local clusters.
- **Exposure-aware**: targets agents under high current or cumulative hazard pressure.
- **Bottleneck-avoidance**: targets active bottleneck zones.

Figure 1 summarizes the policy space. Figure 2 shows where the intervention hook sits in the simulation loop.

3.4 Information-Safety Metrics

The core metric is information-safety efficiency:

$$\eta_{safe} = \frac{\Delta H^+ + \Delta A^+}{1 + P_{exposure} + P_{queue}},$$

where ΔH^+ is positive entropy reduction, ΔA^+ is positive belief-accuracy gain, $P_{exposure}$ is recipient-weighted hazard-load pressure, and P_{queue} is bottleneck pressure observed at intervention time. Chiyoda also records intervention reach, exit imbalance, and a harmful convergence index that rises when entropy reduction coincides with concentrated exit usage, queueing, and exposure.

4 Implementation

Chiyoda is implemented as a Python research toolkit. Scenario YAML files define layouts, cohorts, hazards, responders, information parameters, intervention policies, seeds, and variants. The scenario manager builds

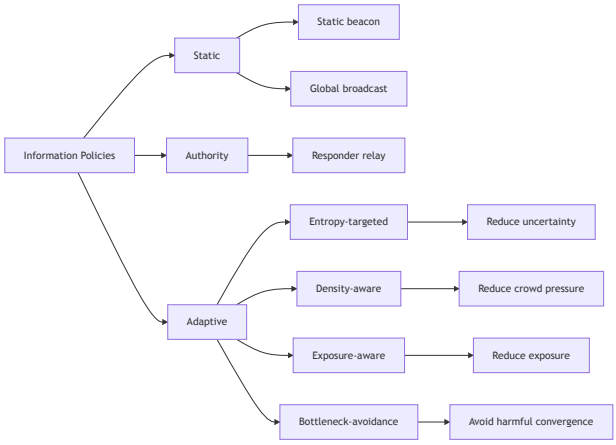

Figure 1: Information intervention policy taxonomy. Policies differ by source, target signal, spatial reach, and safety objective.

Figure 2: Intervention hook in the ITED simulation loop. Policies read the current physical-information state, update recipient beliefs, and emit before/after telemetry before navigation.

a simulation by loading the layout, constructing agents, hazards, exits, responders, navigation, spatial indexing, behavior state machines, and optional intervention policies.

The core implementation lives in `chiyoda/core/simulation.py`. The simulation loop records step telemetry, agent telemetry, cell-level maps, exit flow, bottleneck dynamics, hazard state, gossip events, and intervention events. Study execution lives in `chiyoda/studies/runner.py`; it materializes variants, runs each seed, aggregates tables, and exports study bundles. Analysis and figure export live in `chiyoda/analysis/`.

The information intervention layer lives in `chiyoda/information/interventions.py`. It is deliberately small: policies produce target points and messages, while the shared executor applies messages to recipients and computes before/after entropy and accuracy. This keeps policy design separate from belief mutation and telemetry.

Two implementation details matter for reproducibility. First, A* pathfinding uses per-step caches for density penalties, edge weights, and identical start/goal/belief queries. These caches are cleared whenever the simulation advances to a new navigation phase, so they do not change the cost function; they only avoid recomputing identical values during the same step. Second, gossip reports about the same physical hazard are merged into an existing hazard belief rather than appended as duplicate rumors. This is both a performance requirement and a modeling requirement: repeated reports should update credibility, freshness, severity, and radius estimates for the same hazard, not create an unbounded list of separate hazards.

The paper pipeline mirrors the code pipeline. Running `python3 -m chiyoda.cli sweep scenarios/study_information_control.yaml` emits tables and figures; `paper/scripts/gen_stats.py` converts the exported bundle into `stats.tex`; and the LaTeX build injects those values into the abstract and evaluation.

[TODO] *Replace this implementation sketch with line-referenced architecture details once the model stabilizes.*

5 Evaluation

We evaluate Chiyoda with one primary policy comparison and two supporting stress studies. The primary study is `study_information_control.yaml`, a 50-seed comparison of eight emergency communication regimes: no deliberate intervention, static beacon broadcast, global PA broadcast, responder relay, entropy-targeted communication, density-aware communication, exposure-aware communication, and bottleneck-avoidance communication. The exported bundle contains 400 runs, 8 variant aggregates, and 17780 intervention events. Each run uses the same station-sarin scenario, a 120-agent commuter/tourist population, asym-

metric initial information, gossip, beacon infrastructure, three delayed responders, and a 500-step horizon.

The two supporting studies each use 30 seeds. The intervention ablation study varies targeting budget, timing, and bottleneck awareness. The message-quality study varies broadcast credibility, delay, and repetition. These studies are not replacements for the main comparison; they test whether the main information-safety interpretation is robust to natural design choices.

5.1 Outcome Measures

We report three classes of outcome. Physical outcomes include evacuated agents, mean travel time, incapacitation count, mean hazard exposure, peak bottleneck queue, and exit imbalance. Information outcomes include mean entropy, entropy reduction, intervention reach, entropy reduction attributable to interventions, and belief-accuracy gain. Coupled outcomes include information-safety efficiency (ISE) and harmful convergence index (HCI). These coupled metrics are the main object of the paper: they ask whether a policy buys useful belief improvement without creating exposure or queueing pressure.

5.2 Primary Policy Comparison

Table 1 summarizes the 50-seed variant aggregates. The first result is negative but important: no communication policy dominates the baseline across physical evacuation outcomes. The no-intervention baseline evacuates 10.76 agents on average. Static beacons evacuate 10.90, global broadcast evacuates 11.00, and the remaining active policies range from 9.70 to 10.86. Mann–Whitney tests against the baseline do not show significant improvements in evacuated count or hazard exposure for any active policy at $n = 50$ seeds. This is why the paper cannot claim that more information monotonically improves evacuation.

The second result is that communication policy still matters, even when physical outcomes remain noisy. Static beacons are the best policy by information-safety efficiency (static_beacon, 0.014). They combine local reach, repeated exposure, and high credibility: enough to reduce belief uncertainty without synchronizing the full population toward the same capacity-constrained route. Responder relay has lower efficiency than static beacons, but also the lowest HCI among the active policies in the main study (6.31), which supports the idea that delayed, mobile, high-credibility sources can be conservative safety controllers.

The third result is that wide reach is not the same as useful reach. Global broadcast slightly improves mean evacuation count relative to the baseline (11.00 versus 10.76) and has the lowest mean hazard exposure among the main policies (2.866). However, it ranks fifth by ISE (0.0058) and has a high HCI (8.28). In this scenario, the global message is useful but inefficient: it homogenizes route guidance broadly, while local and capacity-aware policies produce more information benefit per unit of safety pressure.

The fourth result is that adaptive targeting exposes a real control tradeoff. Entropy targeting has the largest intervention accuracy gain in the main study (1.76), and both entropy targeting and bottleneck avoidance significantly reduce mean travel time against no intervention ($p = 0.035$ and $p = 0.014$, respectively). Yet these same policies do not improve evacuation count, and bottleneck avoidance has a near-significant reduction in evacuated agents (-1.06, $p = 0.061$). The practical interpretation is that better-informed trajectories can be shorter for agents who move, while still failing to improve aggregate completion under capacity and hazard pressure.

5.3 Intervention Ablation

Table 2 reports the 30-seed ablation study. The strongest finding is budget non-monotonicity. Early low-budget entropy targeting has the highest ISE in the ablation (0.0253), more than double early high-budget entropy targeting (0.0092). The high-budget version improves belief accuracy more strongly (1.89 versus 1.26), but also produces larger queue pressure and lower information-safety efficiency. This supports the paper's core mechanism: more certainty is not automatically safer when many agents receive similar guidance under shared bottleneck constraints.

The timing ablation also matters. Late entropy targeting recovers evacuation count relative to early entropy targeting (11.47 versus 10.70 under the high-budget setting), but its ISE is lower. Bottleneck targeting is less sensitive to late deployment: the late bottleneck condition has slightly higher evacuation count and slightly higher ISE than early bottleneck targeting. This suggests that bottleneck-aware control is a better candidate for delayed operational response, while entropy targeting needs stricter budget control.

5.4 Message Quality

Table 3 reports the 30-seed message-quality study. Credibility has the expected direction for information efficiency: high-credibility global broadcast has the highest ISE among the broadcast-quality conditions (0.0066), delayed high-credibility broadcast is close behind (0.0063), medium credibility falls to 0.0034, and low credibility falls to 0.0026. Repeating low-credibility messages frequently does not repair this loss; it has the lowest ISE in the study (0.0009) while preserving high harmful convergence (8.92).

This support study strengthens the interpretation of global broadcast. Global messages can increase completed evacuations in some parameterizations, but the information-safety metric penalizes conditions where the same broad signal also creates route synchronization. Message quality therefore behaves like a control parameter, not a cosmetic parameter: credibility and timing change whether communication improves useful beliefs or merely coordinates crowd movement.

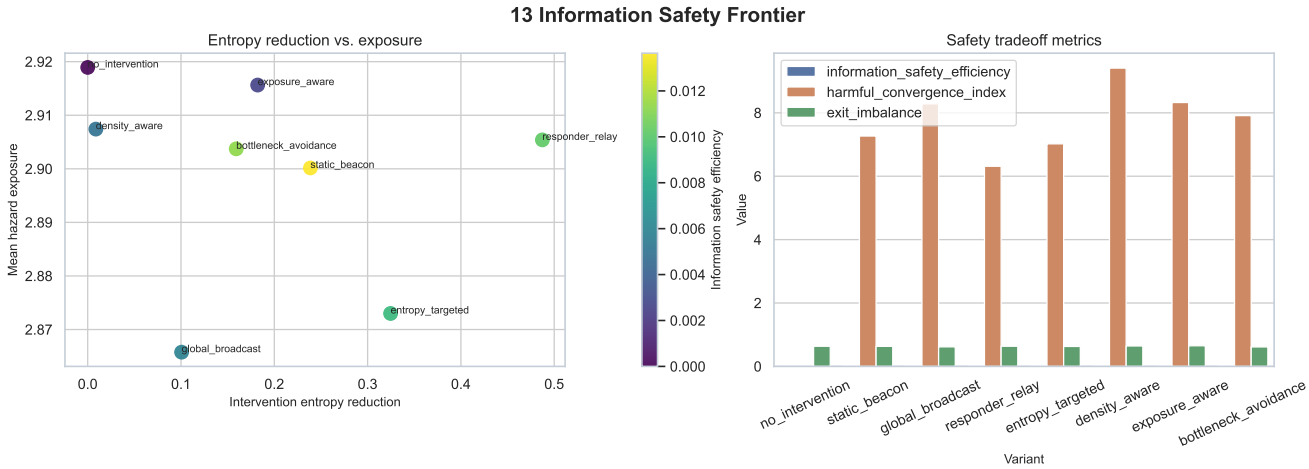

Figure 3: Information-safety frontier across communication policies in the 50-seed main study. Static beacons produce the highest information-safety efficiency, while density-aware and exposure-aware policies show the largest harmful-convergence penalties under this scenario.

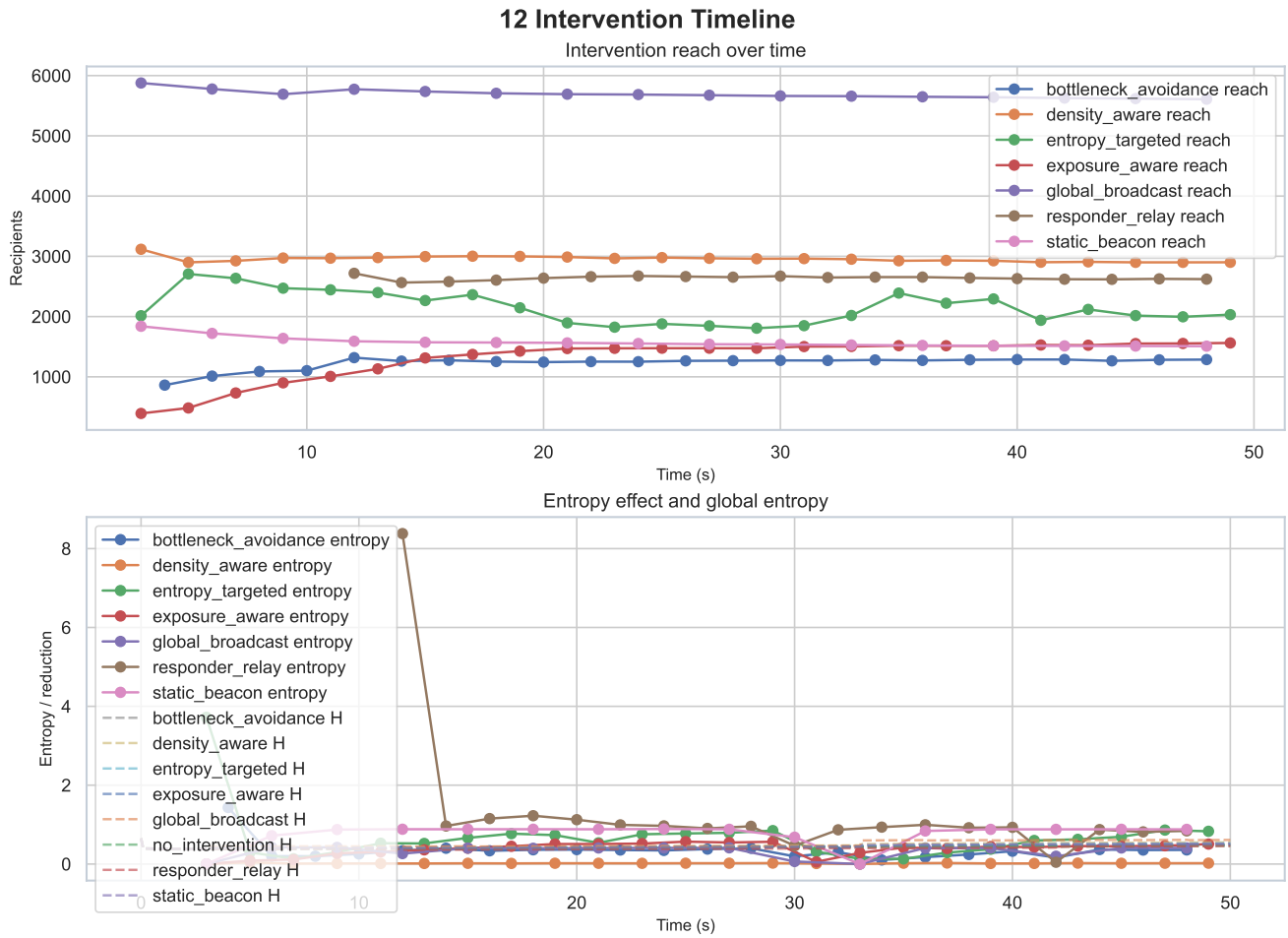

Figure 4: Intervention reach and entropy effects over time. Broad reach and useful safety effect separate: global broadcast reaches many agents, but its information-safety efficiency remains below local static beacons, bottleneck avoidance, responder relay, and entropy targeting.

5.5 Hypotheses Revisited

H1 predicted that adaptive targeting would improve information-safety efficiency over global broadcast. The 50-seed result supports this for static beacons, bottleneck avoidance, responder relay, and entropy targeting, all of which exceed global broadcast by ISE. The result is weaker for density-aware and exposure-aware targeting, which have nonzero efficiency but lower safety efficiency than global broadcast in this scenario.

H2 predicted that global broadcast would reduce entropy quickly but risk route synchronization. The data

Table 1: Variant aggregates from the 50-seed information-control study. ISE denotes information-safety efficiency; HCI denotes harmful-convergence index.

Policy	Evac.	Travel (s)	Exposure	Queue	ISE	HCI
Static beacon	10.90	10.88	2.900	2.72	0.0136	7.27
Bottleneck avoidance	9.70	9.76	2.904	2.84	0.0113	7.91
Responder relay	10.52	10.69	2.905	2.76	0.0102	6.31
Entropy-targeted	9.90	10.31	2.873	2.86	0.0091	7.02
Global broadcast	11.00	11.86	2.866	2.84	0.0058	8.28
Density-aware	10.78	10.63	2.907	2.80	0.0050	9.41
Exposure-aware	10.86	11.65	2.916	2.86	0.0026	8.33
No intervention	10.76	11.53	2.919	2.74	0.0000	0.00

Table 2: Supporting intervention ablation over 30 seeds.

Condition	Evac.	Travel (s)	Exposure	Queue	ISE	HCI
Entropy early, low budget	10.30	11.74	2.809	2.93	0.0253	8.57
Bottleneck late	11.17	11.57	2.873	2.77	0.0143	7.69
Bottleneck early	10.40	10.85	2.778	2.97	0.0140	7.98
Entropy early, high budget	10.70	11.65	2.879	3.00	0.0092	7.08
Entropy late, high budget	11.47	11.80	2.866	2.80	0.0070	7.43
No intervention	11.20	11.84	2.869	2.67	0.0000	0.00

Table 3: Supporting message-quality study over 30 seeds.

Condition	Evac.	Travel (s)	Exposure	Queue	ISE	HCI
High-credibility global	11.13	12.01	2.889	2.83	0.0066	8.42
Delayed high credibility	10.80	10.70	2.856	2.80	0.0063	8.09
Medium-credibility global	11.53	10.85	2.862	2.80	0.0034	7.12
Low-credibility global	12.10	11.24	2.893	2.90	0.0026	9.07
Frequent low credibility	11.93	11.79	2.858	2.80	0.0009	8.92
No intervention	11.20	11.84	2.869	2.67	0.0000	0.00

support the synchronization concern more strongly than the entropy-reduction claim. Global broadcast has high reach and high HCI, but its ISE trails more localized policies. The message-quality study further shows that frequent low-credibility broadcast is especially inefficient.

H3 predicted that exposure-aware targeting would reduce hazard exposure. Exposure-aware targeting does not meaningfully reduce exposure in the 50-seed main study: its mean hazard exposure is 2.916, close to the baseline value of 2.919 and above several other policies. This hypothesis is not supported by the present configuration.

H4 predicted that responder relay would be robust but delayed. The result is partially supported. Responder relay is third by ISE, has the lowest HCI among the main active policies, and avoids the extreme synchronization penalties of some adaptive policies. Its limitation is not obvious physical superiority, but conservative information control.

5.6 Interpretation

The study supports the paper's central framing rather than a single winning policy. Information interventions should be evaluated as safety-control actions, not as monotonic entropy minimizers. A policy can reach many agents and still have low safety efficiency; another can improve belief accuracy and still fail to improve aggregate completion. The promising direction is therefore not "broadcast more," but "broadcast credibly, locally, and with awareness of capacity and exposure." Static beacon and responder relay policies currently look like conservative baselines, while entropy targeting is scientifically valuable because it exposes the failure mode created by overconfident, synchronized route choice.

5.7 Threats to Evaluation

The current evidence is stronger than the pilot run, but still not final external validation. The experiments use one station geometry, one hazard scenario, one population size, and a fixed evacuation horizon. The supporting studies vary intervention design but not architectural layout, sensor latency, or calibrated pedestrian demographics. The HCI and ISE metrics are also model constructs; future work should test alternative normalizations and compare them against trajectory-level bottleneck events, drill data, or expert-coded evacuation plans.

6 Related Work

Chiyoda sits between pedestrian dynamics, hazard-coupled evacuation, and information-aware decision models. Social-force and cellular automata models provide the physical base for movement and congestion [2, 1]. Toxic-gas evacuation work already shows that information perception and transmission affect evacuation outcomes under spreading hazards [4, 3]. Bayesian and game-theoretic models study incomplete information and route choice under congestion [8]. Social-group ABM surveys emphasize communication and collective behavior while documenting gaps in behavioral realism and validation [7, 5].

The distinction is that Chiyoda treats communication as an experimental control surface. It does not only ask whether agents possess information; it asks how an emergency system should allocate limited communication budget under uncertainty. This makes message timing, targeting, credibility, and spatial reach comparable interventions with measurable safety tradeoffs.

[TODO] *Expand this section after the final bibliography pass. In particular, separate empirical evacuation studies, ABM method papers, CBRN-specific evacuation papers, and information/control-theoretic routing papers.*

7 Limitations

Chiyoda is currently a stylized research simulator, not an operational evacuation planner. The station layouts are simplified grids unless imported from richer geometry. Hazard physics are intentionally lightweight and should not be read as validated CBRN dispersion models. Physiological impairment uses piecewise exposure curves that are useful for comparative simulation but not medical prediction.

The behavioral model is also a hypothesis generator rather than a calibrated description of a specific crowd. Agents use simplified BDI-style intentions, local gossip, panic/anxiety states, and familiarity cohorts. These mechanisms are valuable because they make assumptions inspectable, but the paper must not claim real-world predictive validity without calibration against evacuation drills, incident records, controlled VR experiments, or pedestrian trajectory data.

Finally, entropy is not equivalent to safety. The point of the paper is precisely that uncertainty reduction can be harmful when it produces synchronized movement. The proposed information-safety metrics are therefore comparative diagnostics, not universal welfare functions.

8 Conclusion

Chiyoda reframes evacuation as an information-control problem. In a hazard-coupled crowd, emergency communication changes beliefs, beliefs change routes, and route convergence changes exposure and congestion. The framework therefore measures not only whether information spreads, but whether spreading it at a particular time, place, credibility, and radius improves safety.

The repository now contains the code hooks, study definitions, metrics, and paper scaffold needed to evaluate that claim. The next step is empirical: run larger seeded studies, quantify uncertainty, identify regimes where global messages backfire, and refine adaptive policies that reduce entropy without creating harmful convergence.

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